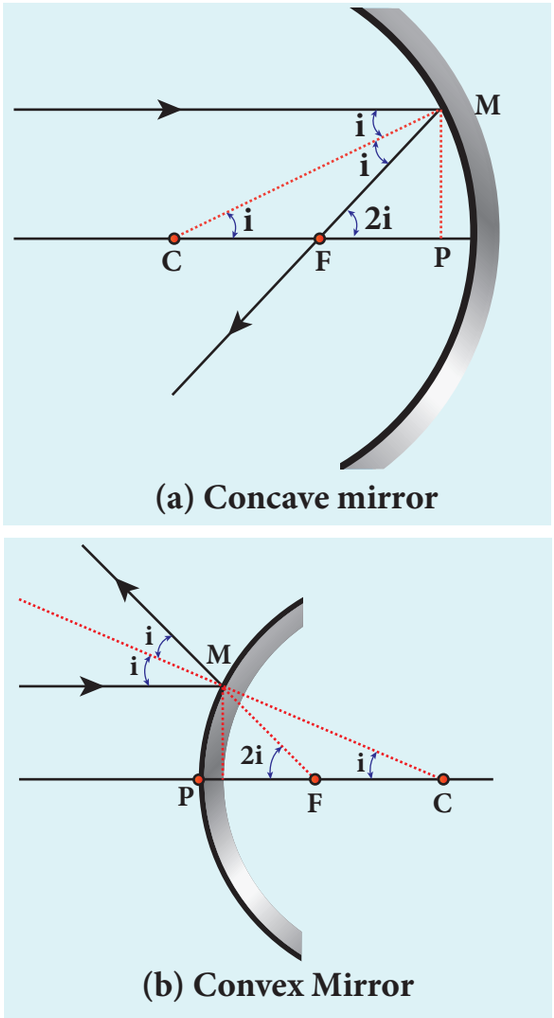


Figure 6.9 Relation between f and R

If MP is the perpendicular from M to the principal axis, then

The angles $\angle MCP = i$ and $\angle MFP = 2i$

From right angle triangles ΔMCP and ΔMFP, we can write,

$$\tan i = \frac{PM}{PC} \text{ and } \tan 2i = \frac{PM}{PF}$$

As the angles are small, $\tan i \approx i$ and $\tan 2i \approx 2i$,

$$i = \frac{PM}{PC} \text{ and } 2i = \frac{PM}{PF}$$

Simplifying further,

$$2\frac{PM}{PC} = \frac{PM}{PF};\ 2PF = PC$$

PF is focal length f and PC is the radius of curvature R.

$$2f = R \qquad \text{(or)} \qquad f = \frac{R}{2} \qquad (6.4)$$

Equation (6.4) is the relation between f and R. The construction is shown for convex mirror in figure 6.9(b)

6.2.3 Image formation in spherical mirrors

The image formed by spherical mirror can be found by ray construction called image tracing. To locate an image point, a minimum of two rays must meet at that point. We can use at least any two of the following four rays as shown in Figure 6.10.

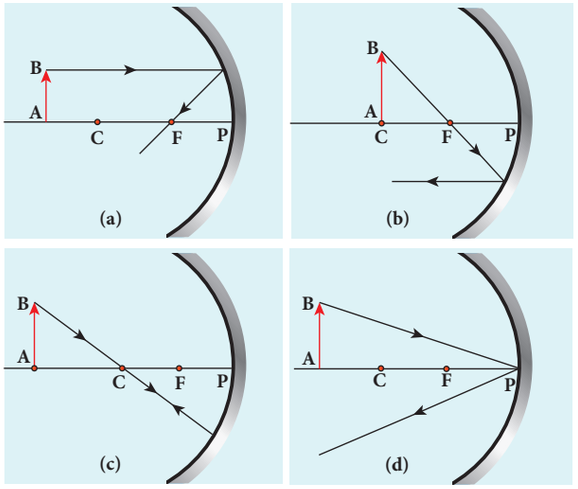


Figure 6.10 Image tracing

(i) A ray parallel to the principal axis after reflection will pass (or appear to pass) through the principal focus. (Figure 6.10(a))

(ii) A ray passing (or appear passing) through the principal focus, after reflection will travel parallel to the principal axis. (Figure 6.10(b))

(iii) A ray passing through the centre of curvature retraces its path after reflection as it is a normal incidence. (Figure 6.10(c))

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